PROJECT PROPOSAL

Group Number (if applicable): _____ **Section:** CSCI - 630.01

Student / Team Member(s):

Junaid Ahmad

Project Title:

Dynamic Search for Pursuit-Evasion Game

Possible Dataset Name / Link (if decided):

What is the problem statement?

To design a framework to analyse different seach algorithms and to explore single vs multi-agent complexity in the context of a Pursuit-Evasion Game.
To find the best path for a dynamically changing destination and start point.

Proposed Methodology

Discuss the types of models or algorithms, validation strategy, evaluation methods, and any major implementation plans.

Maze Generation:
Recursive backtracking
Randomized Prim's Algorithm

Search Algorithms: - Fully Observable
Heuristic based A* search run dynamically to find the target.
MinMax Algorithm to find target before target reaches end of maze.
BFS

I plan on completing these and then further exploring possibilities of adding multiple pursuit agents, changing the possible goal points and exploring how to use the project in Unity3D to make a playable demo.

GitHub Repository Link (Make sure the link is accessible in incognito mode):

https://github.com/Junaidsgs/Dynamic-Search-for-Pursuit-Evasion-Game.git

Note: This proposal is tentative. You may refine or change the project title, dataset, methodology, or scope later based on feedback, further research, or discussion with the instructor. If you need additional space to explain your idea, you may attach an additional page. You may include a block diagram or other visual aids to help convey your project concept.